

Data Analyst Portfolio

By :-

Rohit Sammal



Professional Background

My name is Rohit Sammal from College of Technology, Pantnagar. Currently, I am awaiting my B.Tech-ECE degree. I know several skills including C++, Data Analysis, Python, Arduino IoT Cloud .

I have worked with several companies as an intern like BSNL, Internshala etc. During my internship at Trainity , I gained hands-on experience in data analysis and interpretation, working with real-world datasets to support business decision-making.

Apart from academics , I love playing chess and badminton ; in my leisure time I learn new languages.

As a fresher, I am eager to gain hands-on experience and understand the real challenges of the corporate world. While I have acquired strong theoretical knowledge during my studies, I am keen to apply it in practical scenarios. I consider myself flexible and adaptive, with a strong willingness to learn. I believe that with dedication and consistent effort, I can quickly develop the skills required to contribute effectively and grow within the organization.

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Travel Plan

Plan

Firstly, we decide we want to go on a trip this month.

Prepare

Then, we collect data of various stuff like how many are ready for the trip , when are all available, what will be estimated budget, then route, mode of transportation etc.

Process

Next on , I will make the itinerary to places to visit, list of restaurants, list of hotels, vehicle renting stores, options in mode of transportation etc.

Analyse

Next, we will cancel and add certain locations according to our budget and time constraints and choose most economical and fastest mode of transportation , similarly with other stuff.

Share

Finally, a final plan is shared for more discussion if required and then, everyone agrees for trip.

Act

At last, we finally we start our trip ...

A visit to Doctor's Clinic

Plan

Firstly, a patient suffers from an ailment comes to doctor's clinic.

Prepare

Then, he/she writes his/her problem related data on a clinic paper and rest is told to the doctor about when the problem began , what other problem he/she is facing right now, any past similar experience, any allergy etc.

Process

Next on , Doctor processes given data what could be possible reasons for this problem and ask more question to narrow down.

Analyse

Next, after all discussion doctor on basis of his knowledge and experience decide it is x disease and write few medicines and tell him/her to take certain precautions.

Share

Finally, doctor gives dosage of each medicine and its intake and ask patient for any more queries , if none then return if problem still persists.

Act

At last, patient goes to buy medicines.

Instagram Data Analysis

Overview

In this project, I have analysed Instagram's user data and answer the questions posed by the management team. My insights will help the product manager and the rest of the team make informed decisions about the future direction of the app. My approach will be structured and insight driven.

Approach

Firstly, I thought about main goal of this analysis which is majorly user engagement insights from blocking fake accounts to rewarding top users and I checked who is target audience of my insights , like here it is marketing and investor teams.

Secondly, I created the required dataset from given dataset with commands.

Then, I got familiar with database that is identifying all tables , understanding primary and foreign keys , column meanings and data types etc.

Later, I wrote analytical queries for all questions asked by the teams.

Software

I chose My SQL workbench 8.0 for writing and executing queries, managing schemas as it offers a user friendly, integrated environment for writing SQL queries , database management and visualizing it.

Insights

- Best day to post – Engagement peaked on Thursday.
- Top Hashtags – Smile, beach , party are trending hashtags.
- Bots – There are around 13 potential bots in this given dataset.
- Social Behaviour Analysis – Still , after so many years there are users who have not posted a single pic.

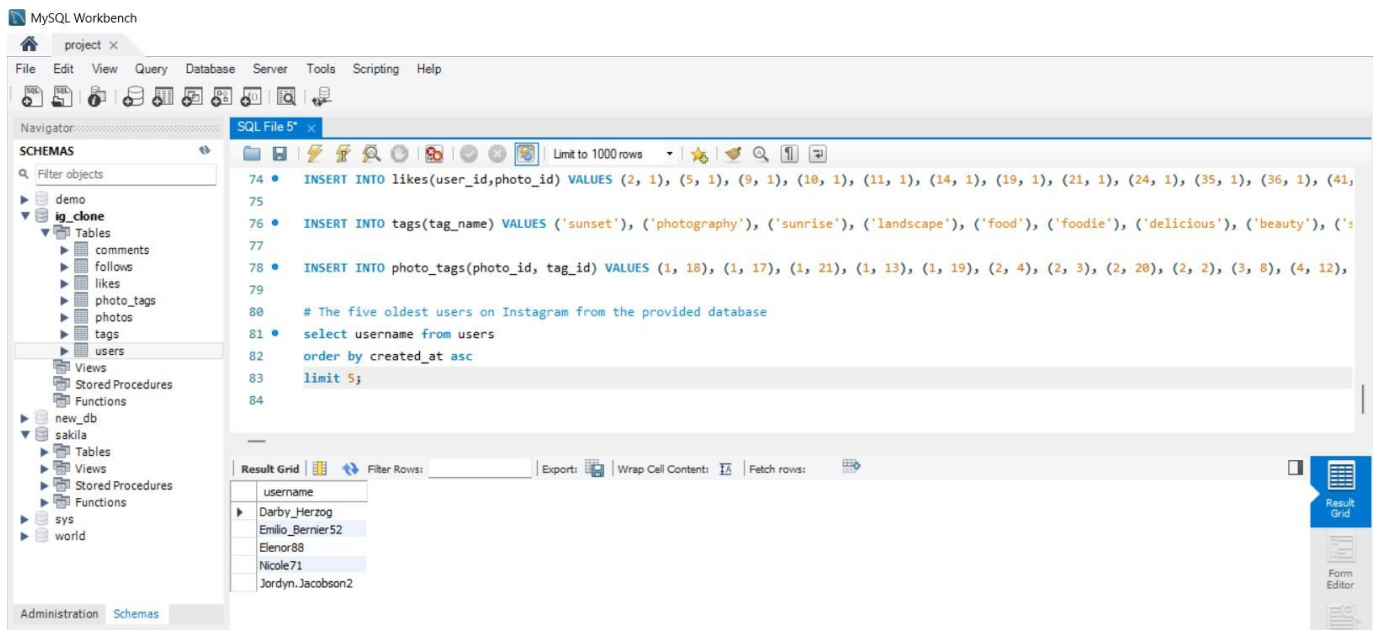
- Activity – the ratio of engagement of active users is around 3.5 vs total users is 2.5 based on posts .

Results

Through this project, I improved my SQL querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends in user engagement and hashtags performance. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

- Loyal User Reward: Top 5 oldest users on the database.



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a project named 'project'. The main editor window shows a SQL file with several queries. The bottom panel displays the 'Result Grid' for the query: `select username from users order by created_at asc limit 5;`. The result grid shows the following usernames:

username
Darby_Herzog
Emilio_Bernier52
Elenor88
Nicole71
Jordyn.Jacobson2

- Inactive Users: Users who never posted a picture.

The screenshot shows the MySQL Workbench interface. On the left, the 'SCHEMAS' pane displays a tree view of databases including 'demo', 'ig_clone', 'new_db', 'sakila', 'sys', and 'world'. The 'demo' database is selected, showing tables like 'comments', 'follows', 'likes', 'photo_tags', 'photos', 'tags', and 'users'. The 'users' table is highlighted, and its structure is shown in the 'Information' pane below: 'Table: users' with columns 'id' (int AI PK), 'username' (varchar(255)), and 'created_at' (timestamp).

The central SQL editor contains the following query:

```

84 # The users who have never posted a single photo on Instagram
85 select u.username,u.id from users u
86 left join
87 photos p on u.id = p.user_id
88 where p.user_id is null;

```

The 'Result Grid' at the bottom displays the query results in a table with two columns: 'username' and 'id'. The results are as follows:

username	id
Aniya_Hackett	5
Kasandra_Homenick	7
Jadyn81	14
Rocio33	21
Maxwell_Halvorson	24
Tierra_Trantow	25
Pearl7	34
Ollie_Ledner37	36
Mckenna17	41
David_Osinski47	45
Morgan_Kassulke	49
Linnea59	53
Duane60	54
Julien_Schmidt	57
Mike_Auer39	66
Franco_Keebler64	68
Nia_Haag	71
Hulda_Macejkovic	74
Leslie67	75
Janelle_Nikolaus81	76
Darby_Herzog	80
Esther_Zulauf61	81
Bartholome_Bernhard	83
Jessyca_West	89
Esmeralda_Mraz57	90
Bethany20	91

- Contest Winner: Zack's picture got most likes.

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigators

SCHEMAS

Filter objects

demo

ig_clone

Tables

comments

follows

likes

photo_tags

photos

tags

users

Views

Stored Procedures

Functions

new_db

sakila

Tables

Views

Stored Procedures

Functions

sys

world

Administration Schemas Information

Trinity x

Limit to 1000 rows

```
88 where p.user_id is null;
89 # The user with the most likes on a single photo wins
90 • select id,username from users
91     where id = (
92         select user_id from photos
93         where id = (
94             select photo_id from likes
95             group by photo_id
96             order by count(*) desc
97             limit 1
98         )
99     );
```

Result Grid

id	username
52	Zack_Kemmer93
NULL	NULL

- Trending Hashtags

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigators

SCHEMAS

Filter objects

demo

ig_clone

Tables

comments

follows

likes

photo_tags

photos

tags

users

Views

Stored Procedures

Functions

new_db

sakila

Tables

Views

Stored Procedures

Functions

sys

world

Administration Schemas Information

Trinity x

Limit to 1000 rows

```
98 )
99 );
100
101 # Identify the top five most commonly used hashtags on the platform
102 • select t.tag_name as top_tags, count(p.tag_id) as use_count
103     from tags t
104     join photo_tags p on p.tag_id = t.id
105     group by p.tag_id
106     order by use_count desc
107     limit 5;
108
109
```

Result Grid

top_tags	use_count
smile	59
beach	42
party	39
fun	38
concert	24

- Highest registrations day – Thursday

MySQL Workbench interface showing two SQL queries and their results.

Query 1:

```

102 • select t.tag_name as top_tags, count(p.tag_id) as use_count
103       from tags t
104       join photo_tags p on p.tag_id = t.id
105       group by p.tag_id
106       order by use_count desc
107       limit 5;
108 # The day of the week when most users register on Instagram
109 • select count(dayname(created_at)) as day_count , dayname(created_at) as day_name from users
110       group by day_name
111       order by day_count desc
112       limit 1;
113
114
115

```

Result Grid:

day_name	day_count
Thursday	16

Table: tags

- User engagement : the average number of posts per user on Instagram.

MySQL Workbench interface showing two SQL queries and their results.

Query 1:

```

103       from tags t
104       join photo_tags p on p.tag_id = t.id
105       group by p.tag_id
106       order by use_count desc
107       limit 5;
108 # The day of the week when most users register on Instagram
109 • select count(*) as day_count , dayname(created_at) as day_name from users
110       group by day_name
111       order by day_count desc
112       limit 1;
113 # the average number of posts per user and the total number of photos on Instagram divided by the total number of users
114 • select count(p.id)/count(distinct(p.user_id)) as avg_active_users , count(p.id) / count(distinct (u.id)) as avg_total_users
115       from users u left join photos p on u.id = p.user_id ;
116

```

Result Grid:

avg_active_users	avg_total_users
3.4730	2.5700

Table: tags

- List of potential bots

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

demo

ig_clone

Tables

comments

follows

likes

photo_tags

photos

tags

users

Views

Stored Procedures

Functions

new_db

sakila

Tables

Views

Stored Procedures

Functions

sys

world

Administration Schemas

Information

Table: tags

Columns:

id int AI PK

tag_name varchar(255)

created_at timestamp

Trinity*

Limit to 1000 rows

```
118 # The users who have liked every single photo on the site
119 • select l.user_id,u.username, count(distinct(l.photo_id)) as total_pics_liked
120 from likes l
121 join users u on u.id = l.user_id
122 group by l.user_id
123 having total_pics_liked = max(distinct (l.photo_id)) ;
124
125
126
127
128
```

Result Grid

Filter Rows:

Export: Wrap Cell Contents

user_id	username	total_pics_liked
5	Aniya_Hackett	257
14	Jadyn81	257
21	Roco33	257
24	Maxwell.Halvorson	257
36	Ollie_Ledner37	257
41	Mckenna17	257
54	Duane60	257
57	Julien_Schmidt	257
66	Mike_Auer39	257
71	Nia_Haag	257
75	Leslie67	257
76	Janelle.Nikolaus81	257
91	Bethany20	257

Form Editor

Field Types

Query Stats

Project 3 - Operation Analytics and Investigating Metric Spike

Overview

In this project, I was provided with various datasets and tables, and I derived insights from this data to answer questions posed by different departments within the company. I used my advanced SQL skills to analyse the data and provide valuable insights that can help improve the company's operations and understand sudden changes in key metrics. My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings and data types etc... , exploring dimensions, timestamps etc. then thought about main goal of this analysis which is majorly about user engagement on weekly basis for products and emails , job reviews , language used metrics.

Secondly, I created the required dataset from given dataset with commands. Later, I wrote analytical queries for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

I chose My SQL workbench 8.0 for writing and executing queries, managing

schemas as it offers a user friendly, integrated environment for writing SQL queries , database management and visualizing it.

Insights

- **Jobs Reviewed Over Time:** job review were low but time spent remained constant.
- **Throughput Analysis:** weekly rollout is better since it smooths out daily fluctuations.
- **Language Share Analysis:** French is most used language .
- **Weekly User Engagement:** engagement of users has increased over weeks but not significantly.
- **User Growth Analysis :** product usage has increased significantly over week.
- **Weekly Retention Analysis:** after signing up product, most users are engaging with product.
- **Email Engagement Analysis:** this is also in rise with each week.

Results

Through this project, I improved my SQL querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends in user engagement, different languages, and devices usage. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

- **Job reviewed over time:**

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator: Project3 x

Limit to 1000 rows

106

107 • `select * from job ;`

108

109 `# An SQL query to calculate the number of jobs reviewed per hour`

110

111 • `select ds, count(*) * 3600/sum(time_spent) as jobs_per_hour from job`

112 `where event = 'decision'`

113 `group by ds`

114 `order by ds ;`

115

116

117

118

119

120

Result Grid

ds	jobs_per_hour
2020-11-21	37.1134
2020-11-24	42.8571
2020-11-27	34.6154
2020-11-28	327.2727
2020-11-29	180.0000

Administration Schemas

Information

Table: job

Columns:

- Throughput analysis:

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator: Project3* x

Limit to 1000 rows

113 `group by ds`

114 `order by ds ;`

115

116 `# An SQL query to calculate the 7-day rolling average of throughput`

117 • `WITH daily_throughput as (`

118 `select ds, count(*)/sum(time_spent) as throughput from job`

119 `group by ds )`

120 `select ds, throughput, avg(throughput) over ( order by ds`

121 `) as throughput_7d from daily_throughput`

122 `where ds >= '2020-11-24'`

123 `order by ds;`

124

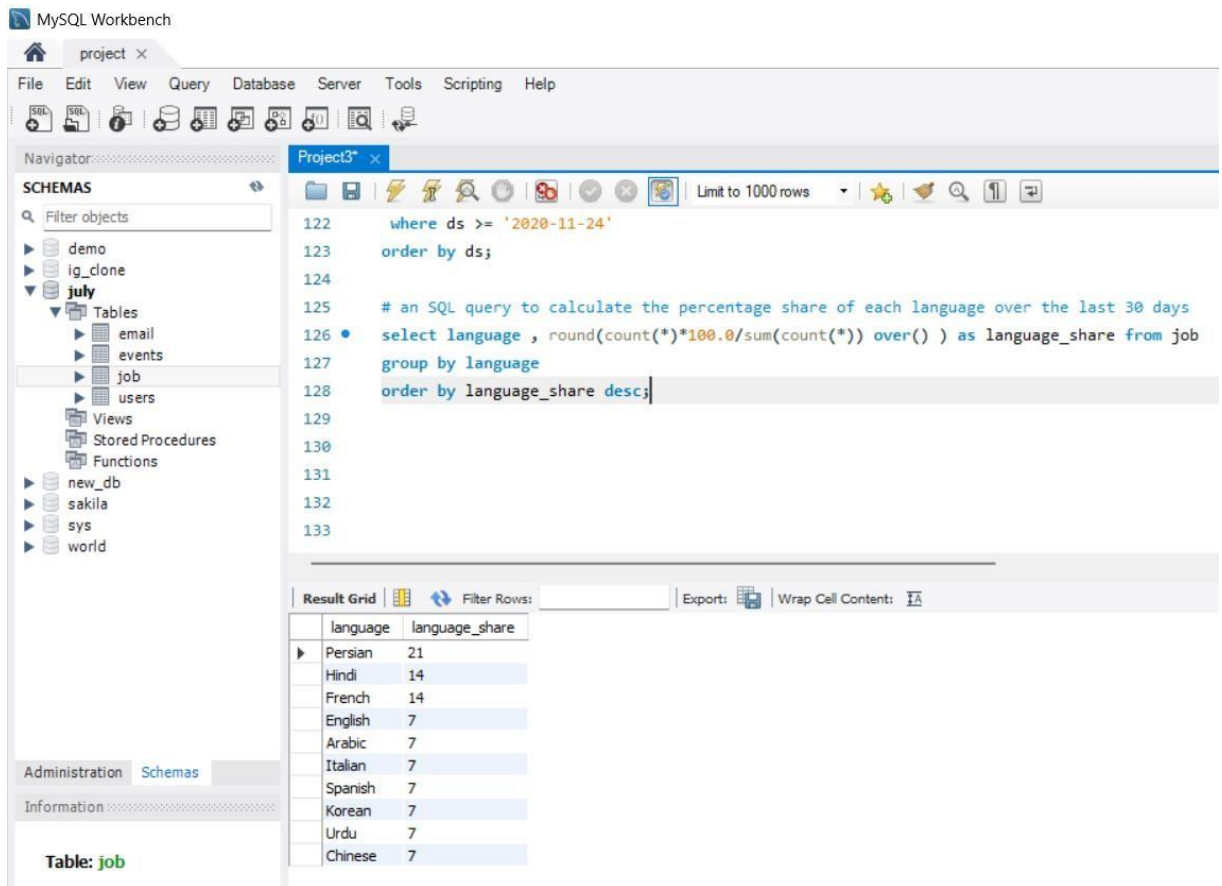
Result Grid

ds	throughput	throughput_7d
2020-11-24	0.0119	0.01190000
2020-11-25	0.0222	0.01705000
2020-11-26	0.0179	0.01733333
2020-11-27	0.0096	0.01540000
2020-11-28	0.0606	0.02444000
2020-11-29	0.0500	0.02870000
2020-11-30	0.0500	0.03174286

Administration Schemas

Information

- **Language Share Analysis:**



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'job' table selected under the 'july' database. The main editor window contains the following SQL query:

```

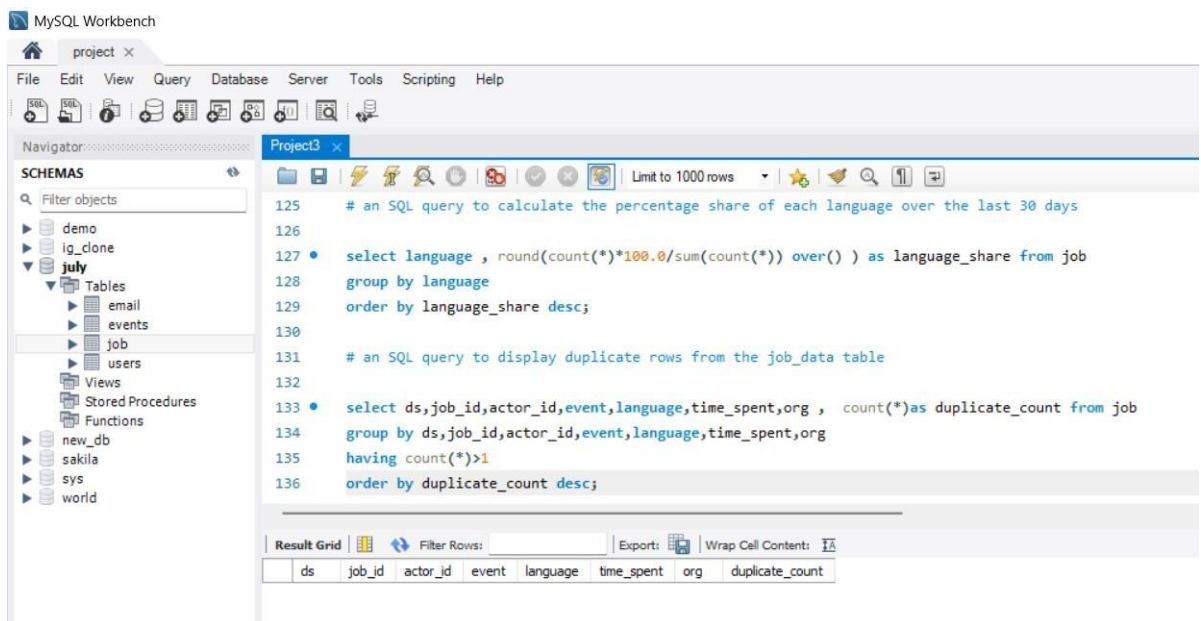
122     where ds >= '2020-11-24'
123     order by ds;
124
125     # an SQL query to calculate the percentage share of each language over the last 30 days
126 •   select language , round(count(*)*100.0/sum(count(*) over() ) as language_share from job
127     group by language
128     order by language_share desc;
129
130
131
132
133

```

Below the query editor, the 'Result Grid' tab is active, displaying the following data:

language	language_share
Persian	21
Hindi	14
French	14
English	7
Arabic	7
Italian	7
Spanish	7
Korean	7
Urdu	7
Chinese	7

- **Duplicate Rows Detection:**



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with the 'job' table selected under the 'july' database. The main editor window contains the following SQL queries:

```

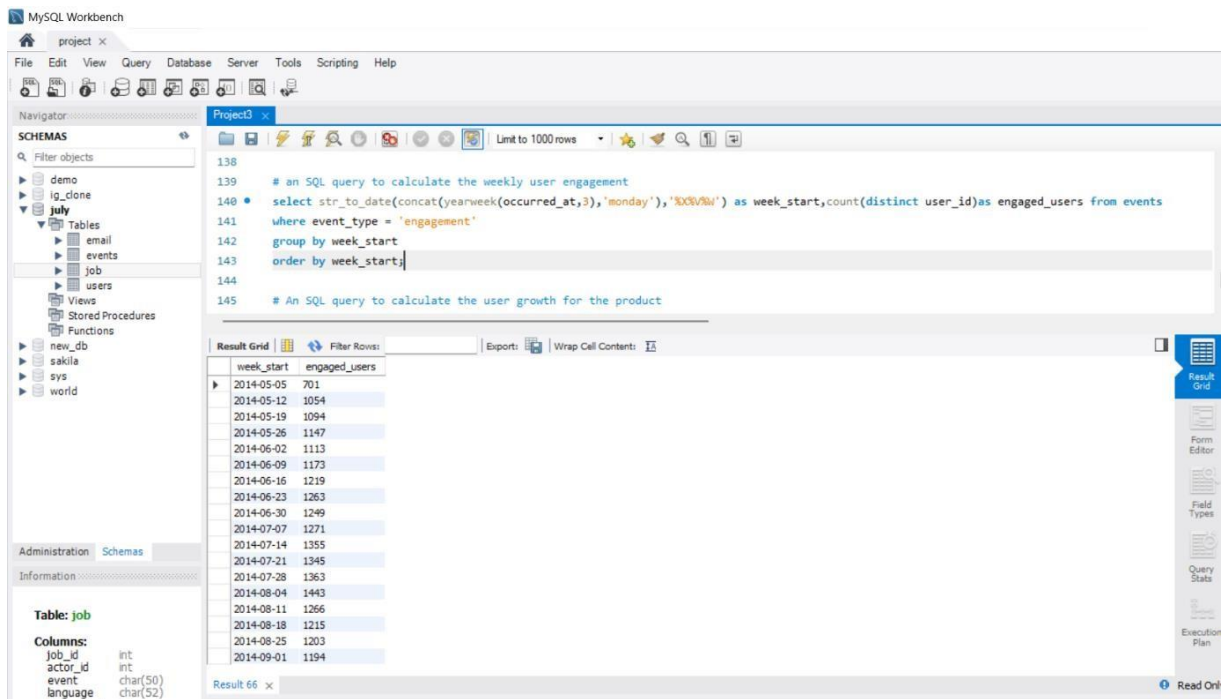
125     # an SQL query to calculate the percentage share of each language over the last 30 days
126
127 •   select language , round(count(*)*100.0/sum(count(*) over() ) as language_share from job
128     group by language
129     order by language_share desc;
130
131     # an SQL query to display duplicate rows from the job_data table
132
133 •   select ds,job_id,actor_id,event,language,time_spent,org , count(*)as duplicate_count from job
134     group by ds,job_id,actor_id,event,language,time_spent,org
135     having count(*)>1
136     order by duplicate_count desc;

```

Below the queries, the 'Result Grid' tab is active, displaying the following data:

ds	job_id	actor_id	event	language	time_spent	org	duplicate_count
----	--------	----------	-------	----------	------------	-----	-----------------

- Weekly User Engagement:



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a tree view expanded to 'July' and 'Users'. The main editor window contains the following SQL query:

```

138
139 # An SQL query to calculate the weekly user engagement
140 • select str_to_date(concat(yearweek(occurred_at,3),'monday'), '%X%V%W') as week_start, count(distinct user_id) as engaged_users from events
141 where event_type = 'engagement'
142 group by week_start
143 order by week_start;
144
145 # An SQL query to calculate the user growth for the product
  
```

The 'Result Grid' at the bottom shows the following data:

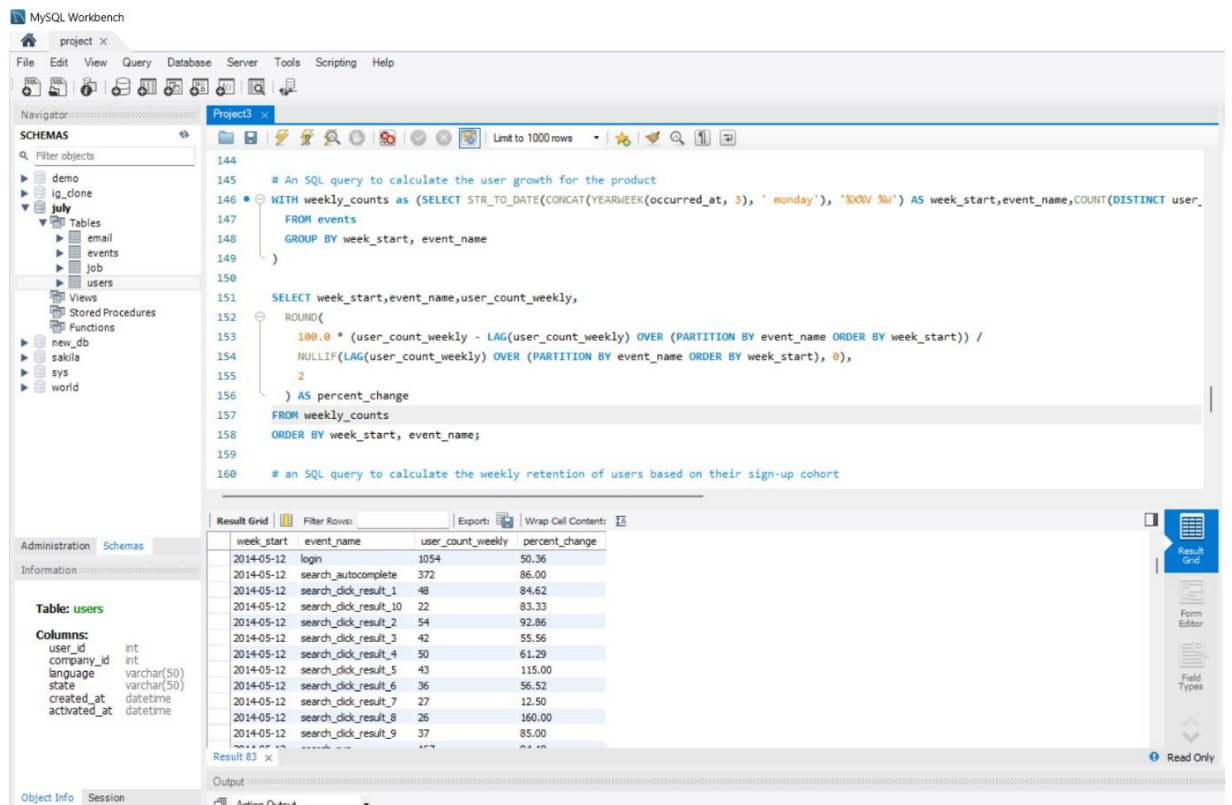
week_start	engaged_users
2014-05-05	701
2014-05-12	1054
2014-05-19	1094
2014-05-26	1147
2014-06-02	1113
2014-06-09	1173
2014-06-16	1219
2014-06-23	1263
2014-06-30	1249
2014-07-07	1271
2014-07-14	1355
2014-07-21	1345
2014-07-28	1363
2014-08-04	1443
2014-08-11	1266
2014-08-18	1215
2014-08-25	1203
2014-09-01	1194

The 'Table: job' information is shown in the bottom left:

```

Columns:
job_id      int
actor_id   int
event       char(50)
language    char(52)
  
```

- User Growth Analysis:



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with a tree view expanded to 'July' and 'Users'. The main editor window contains the following SQL query:

```

144
145 # An SQL query to calculate the user growth for the product
146 • WITH weekly_counts as (SELECT STR_TO_DATE(CONCAT(YEARWEEK(occurred_at, 3), 'monday'), '%X%V %W') AS week_start, event_name, COUNT(DISTINCT user_
147 FROM events
148 GROUP BY week_start, event_name
149 )
150
151 SELECT week_start, event_name, user_count_weekly,
152 ROUND(
153 100.0 * (user_count_weekly - LAG(user_count_weekly) OVER (PARTITION BY event_name ORDER BY week_start)) /
154 NULLIF(LAG(user_count_weekly) OVER (PARTITION BY event_name ORDER BY week_start), 0),
155 2
156 ) AS percent_change
157 FROM weekly_counts
158 ORDER BY week_start, event_name;
159
160 # An SQL query to calculate the weekly retention of users based on their sign-up cohort
  
```

The 'Result Grid' at the bottom shows the following data:

week_start	event_name	user_count_weekly	percent_change
2014-05-12	login	1054	50.36
2014-05-12	search_autocomplete	372	86.00
2014-05-12	search_click_result_1	48	84.62
2014-05-12	search_click_result_10	22	83.33
2014-05-12	search_click_result_2	54	92.86
2014-05-12	search_click_result_3	42	55.56
2014-05-12	search_click_result_4	50	61.29
2014-05-12	search_click_result_5	43	115.00
2014-05-12	search_click_result_6	36	56.52
2014-05-12	search_click_result_7	27	12.50
2014-05-12	search_click_result_8	26	160.00
2014-05-12	search_click_result_9	37	85.00

The 'Table: users' information is shown in the bottom left:

```

Columns:
user_id      int
company_id  int
language     varchar(50)
state        varchar(50)
created_at   datetime
activated_at datetime
  
```

- Weekly Retention Analysis:

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator: Project3 x

Limit to 1000 rows

151 # an SQL query to calculate the weekly retention of users based on their sign-up cohort

152 WITH signed_up_users AS (

153 SELECT DISTINCT user_id

154 FROM events

155 WHERE event_type = 'signup_flow'

156),

157

158 weekly_engaged_users AS (

159 SELECT

160 STR_TO_DATE(CONCAT(YEARWEEK(occurred_at, 3), ' Monday'), '%X%V %W') AS week_start,

161 user_id

162 FROM events

163 WHERE event_type = 'engagement'

164)

165

166 SELECT

167 week_start,

168 COUNT(DISTINCT w.user_id) AS active_users_after_signup

169 FROM weekly_engaged_users w

170 JOIN signed_up_users s ON w.user_id = s.user_id

171 GROUP BY week_start

172 ORDER BY week_start;

Administration Schemas

Information

Table: users

Columns:

user_id int

company_id int

language varchar(50)

state varchar(50)

created_at datetime

activated_at datetime

Result Grid

Filter Rows:

Export: Wrap Cell Content: 1A

week_start	active_users_after_signup
2014-05-05	81
2014-05-12	224
2014-05-19	317
2014-05-26	407
2014-06-02	444
2014-06-09	510
2014-06-16	534

Result 69 x

- **Weekly Engagement Per Device:**

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator

Project3 x

Limit to 1000 rows

175 # an SQL query to calculate the weekly engagement per device

176 • SELECT STR_TO_DATE(CONCAT(YEARWEEK(occurred_at, 3), ' Monday'), '%X%W %u') AS week_start, device, count(distinct user_id) as engaged_users

177 FROM events

178 WHERE event_type = 'engagement'

179 GROUP BY week_start, device

180 ORDER BY week_start, device;

Result Grid

week_start	device	engaged_users
2014-05-05	acer aspire desktop	10
2014-05-05	acer aspire notebook	21
2014-05-05	amazon fire phone	4
2014-05-05	asus chromebook	23
2014-05-05	dell inspiron desktop	21
2014-05-05	dell inspiron notebook	49
2014-05-05	hp pavilion desktop	15
2014-05-05	htc one	16
2014-05-05	ipad air	30
2014-05-05	ipad mini	21
2014-05-05	iphone 4s	21
2014-05-05	iphone 5	70
2014-05-05	iphone 5s	45
2014-05-05	kindle fire	6
2014-05-05	lenovo thinkpad	90
2014-05-05	mac mini	8
2014-05-05	macbook air	57
2014-05-05	macbook pro	154
2014-05-05	nexus 10	16
2014-05-05	nexus 5	43
2014-05-05	nexus 7	20
2014-05-05	nokia lumia 635	19
2014-05-05	samsung galaxy tablet	8
2014-05-05	samsung galaxy note	7
2014-05-05	samsung galaxy s4	56
2014-05-05	windows surface	10
2014-05-12	acer aspire desktop	26

Result 74 x

Output

Object Info Session

Table: users

Columns:

- user_id int
- company_id int
- language varchar(50)
- state varchar(50)
- created_at datetime
- activated_at datetime

- **Email Engagement Analysis:**

MySQL Workbench

project x

File Edit View Query Database Server Tools Scripting Help

Navigator

Project3 x

Limit to 1000 rows

183 # an SQL query to calculate the email engagement metrics

184 • SELECT STR_TO_DATE(CONCAT(YEARWEEK(occured_at, 3), ' Monday'), '%X%W %u') AS week_start, action, COUNT(*) AS action_count

185 FROM email

186 GROUP BY week_start, action

187 ORDER BY week_start, action;

188

189

Result Grid

week_start	action	action_count	user_count
2014-05-05	email_clickthrough	187	187
2014-05-05	email_open	332	332
2014-05-05	sent_reengagement_email	98	98
2014-05-05	sent_weekly_digest	908	908
2014-05-12	email_clickthrough	434	431
2014-05-12	email_open	919	910
2014-05-12	sent_reengagement_email	164	164
2014-05-12	sent_weekly_digest	2602	2602
2014-05-19	email_clickthrough	479	478
2014-05-19	email_open	971	961
2014-05-19	sent_reengagement_email	175	175
2014-05-19	sent_weekly_digest	2665	2665
2014-05-26	email_clickthrough	498	495
2014-05-26	email_open	995	984
2014-05-26	sent_reengagement_email	179	179
2014-05-26	sent_weekly_digest	2733	2733
2014-06-02	email_clickthrough	453	447
2014-06-02	email_open	1026	1008
2014-06-02	sent_reengagement_email	179	179
2014-06-02	sent_weekly_digest	2822	2822
2014-06-09	email_clickthrough	492	483

Result 80 x

Output

Table: users

Columns:

- user_id int
- company_id int
- language varchar(50)
- state varchar(50)
- created_at datetime
- activated_at datetime

Project 4- Hiring Process Analytics

Overview

In this project, as a data analyst, I am given a dataset containing records of previous hires. My job is to analyze this data and answer certain questions that can help the company improve its hiring process.

Here's what I'll be doing:

1. **Handling Missing Data:** Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. **Clubbing Columns:** If there are columns with multiple categories that can be combined, do so to simplify your analysis.
3. **Outlier Detection:** Check for outliers in the dataset that may skew your analysis.
4. **Removing Outliers:** Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. **Data Summary:** After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings and data types etc... , exploring dimensions, timestamps etc. then thought about main goal of this analysis which is majorly about hiring and department wise analysis.

Secondly, I created the required dataset from given dataset using various excel functions.

Later, I wrote excel commands for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

Insights

- **Hiring Analysis:** Males : Females hired ratio is 1.38:1
- **Salary Analysis:** Average salary of general management is highest and marketing department is lowest as compared to other departments and overall.
- **Salary Distribution:** Most employees' salary range from 40000 to 80000.
- **Departmental Analysis:** Operations department has highest employees hired.
- **Position Tier Analysis:** c9 post has highest number of employees

Results

Through this project, I improved my Excel querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends in hiring in different gender, and position tiers. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

- **Hiring Analysis:**

=COUNTIFS(\$D\$2:\$D\$7169,"Male",\$C\$2:\$C\$7169,"Hired")														
Application_id	Interview Taken on	Status	Event_name	Department	Post Name	Offered Salary								
383422	5/1/14 11:40	Hired	Male	Service Department	c8	56553								
907518	5/6/14 8:08	Hired	Female	Service Department	c5	22075				Gender	Total Hired			
176719	5/6/14 8:08	Rejected	Male	Service Department	c5	70069				Male =	2563			
429799	5/2/14 16:28	Rejected	Female	Operations Department	i4	3207				Female=	1856			
253651	5/2/14 16:32	Hired	Male	Operations Department	i4	29668				Null=	10			
289907	5/1/14 7:44	Hired	Male	Sales Department	-	85914				Don't want to say=	268			
959124	5/6/14 16:27	Rejected	Male	Sales Department	i7	69904								
86642	5/9/14 13:17	Rejected	Male	Sales Department	i7	11758								
751029	5/2/14 13:09	Hired	Female	Service Department	i4	15156								
434547	5/2/14 13:11	Rejected	Female	Service Department	i4	49515								
518854	5/1/14 9:00	Rejected	Male	Service Department	n10	26990								

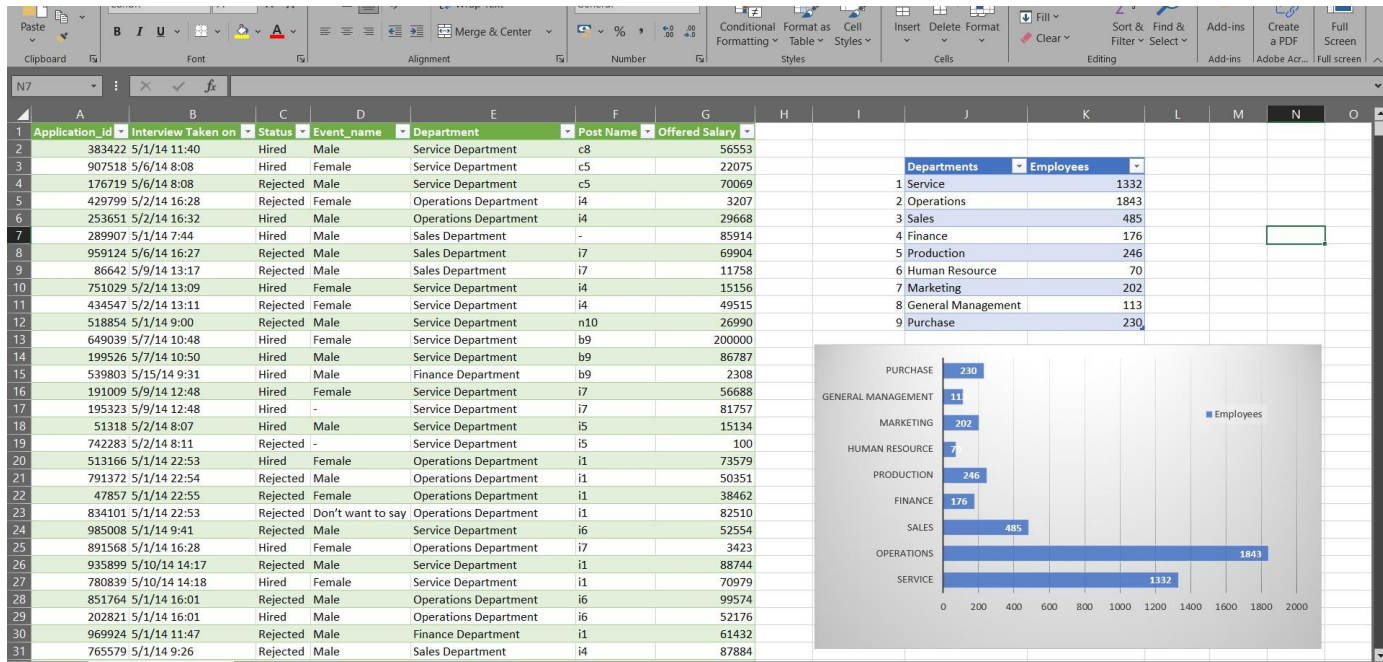
- Salary analysis:

=ROUND(AVERAGEIF(\$C\$2:\$C\$7169,\$C\$2,\$G\$2:\$G\$7169),0)														
Application_id	Interview Taken on	Status	Event_name	Department	Post Name	Offered Salary								
383422	5/1/14 11:40	Hired	Male	Service Department	c8	56553				Average Salary=	49753			
907518	5/6/14 8:08	Hired	Female	Service Department	c5	22075				Department wise	Average Salary			
176719	5/6/14 8:08	Rejected	Male	Service Department	c5	70069				Service	50550			
429799	5/2/14 16:28	Rejected	Female	Operations Department	i4	3207				Sales	48540			
253651	5/2/14 16:32	Hired	Male	Operations Department	i4	29668				Finance	48748			
289907	5/1/14 7:44	Hired	Male	Sales Department	-	85914				Operations	48914			
959124	5/6/14 16:27	Rejected	Male	Sales Department	i7	69904				General Management	60810			
86642	5/9/14 13:17	Rejected	Male	Sales Department	i7	11758				Production	49351			
751029	5/2/14 13:09	Hired	Female	Service Department	i4	15156				Human Resource	49014			
434547	5/2/14 13:11	Rejected	Female	Service Department	i4	49515				Marketing	47843			
518854	5/1/14 9:00	Rejected	Male	Service Department	n10	26990				Purchase	52087			
649039	5/7/14 10:48	Hired	Female	Service Department	b9	200000								
199526	5/7/14 10:50	Hired	Male	Service Department	b9	86787								
539803	5/15/14 9:31	Hired	Male	Finance Department	b9	2308								
191009	5/9/14 12:48	Hired	Female	Service Department	i7	56688								

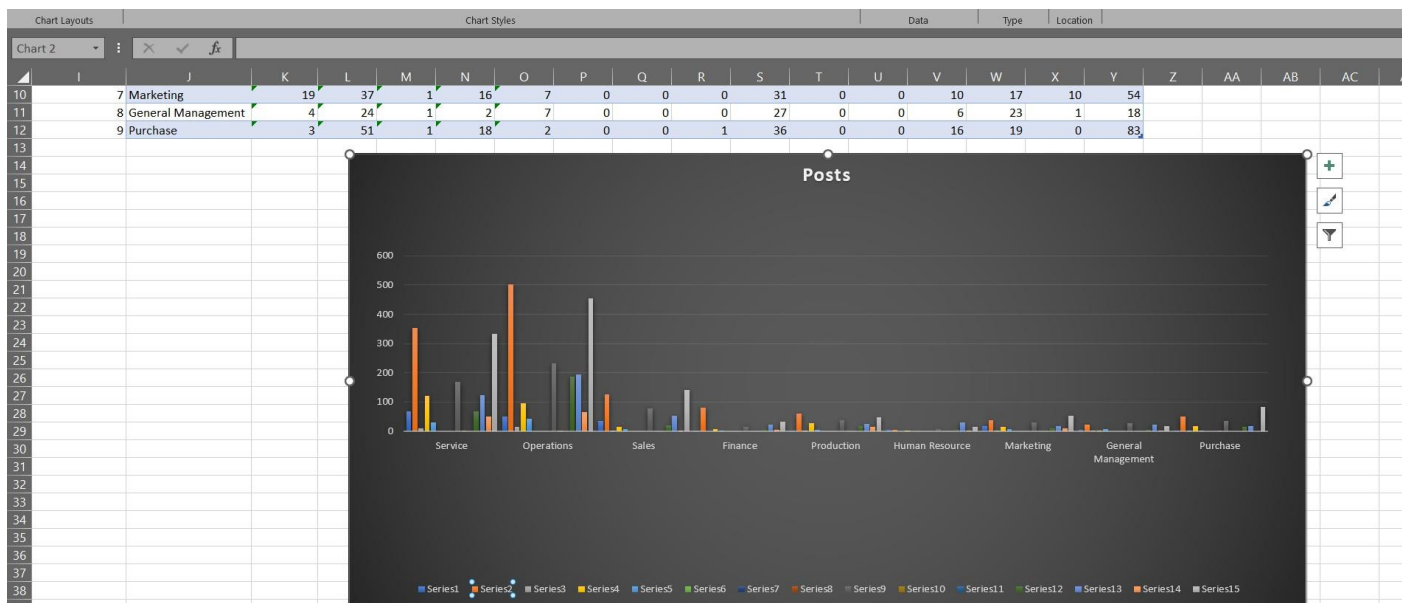
- Salary Distribution :

=COUNTIFS(\$C\$2:\$C\$7169,\$C\$2,\$G\$2:\$G\$7169,">="&J5,\$G\$2:\$G\$7169,"<="&K5)														
Application_id	Interview Taken on	Status	Event_name	Department	Post Name	Offered Salary								
383422	5/1/14 11:40	Hired	Male	Service Department	c8	56553				Class width =	39920			
907518	5/6/14 8:08	Hired	Female	Service Department	c5	22075				Class Intervals	Salary From	To	Frequency	
176719	5/6/14 8:08	Rejected	Male	Service Department	c5	70069					800	40720	1910	
429799	5/2/14 16:28	Rejected	Female	Operations Department	i4	3207					40720	80640	1943	
253651	5/2/14 16:32	Hired	Male	Operations Department	i4	29668					80640	120560	841	
289907	5/1/14 7:44	Hired	Male	Sales Department	-	85914					120560	160480	0	
959124	5/6/14 16:27	Rejected	Male	Sales Department	i7	69904					160480	200400	1	
86642	5/9/14 13:17	Rejected	Male	Sales Department	i7	11758					200400	240320	0	
751029	5/2/14 13:09	Hired	Female	Service Department	i4	15156					240320	280240	0	
434547	5/2/14 13:11	Rejected	Female	Service Department	i4	49515					280240	320160	1	
518854	5/1/14 9:00	Rejected	Male	Service Department	n10	26990					320160	360080	0	
649039	5/7/14 10:48	Hired	Female	Service Department	b9	200000					360080	400000	1	
199526	5/7/14 10:50	Hired	Male	Service Department	b9	86787								
539803	5/15/14 9:31	Hired	Male	Finance Department	b9	2308								

- Departmental Analysis:



Position Tier Analysis:



Project 5- IMDB Movie Analysis

Overview

In this project, as a data analyst, I am given a dataset related to IMDB movies. My job is to analyse this data and answer certain questions that can help stakeholders make informed decisions.

Here's what I'll be doing:

1. **Handling Missing Data:** Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. **Clubbing Columns:** If there are columns with multiple categories that can be combined, do so to simplify your analysis.
3. **Outlier Detection:** Check for outliers in the dataset that may skew your analysis.
4. **Removing Outliers:** Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. **Data Summary:** After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings and data types etc... , exploring dimensions, budget,ratings etc. then thought about main goal of this analysis which is majorly around IMDB ratings and net profit.

Secondly, I created the required dataset from given dataset using various excel functions.

Later, I wrote excel commands for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

Insights

- **Movie Genre Analysis:** Drama, Comedy, Thriller movies are produced in large proportion but have average rating of 6.6 .
- **Movie Duration Analysis:** Mostly movies with 110 to 150 minutes duration have high ratings.
- **Language Analysis:** While English movies are in huge amount but Japanese movies are once with high ratings.
- **Director Analysis:** Director with above average 8.2 IMDB rating scores are in Top 1%.
- **Budget Analysis:** Good will is highest profitable movies and budget and gross collection have weak relationship.

Results

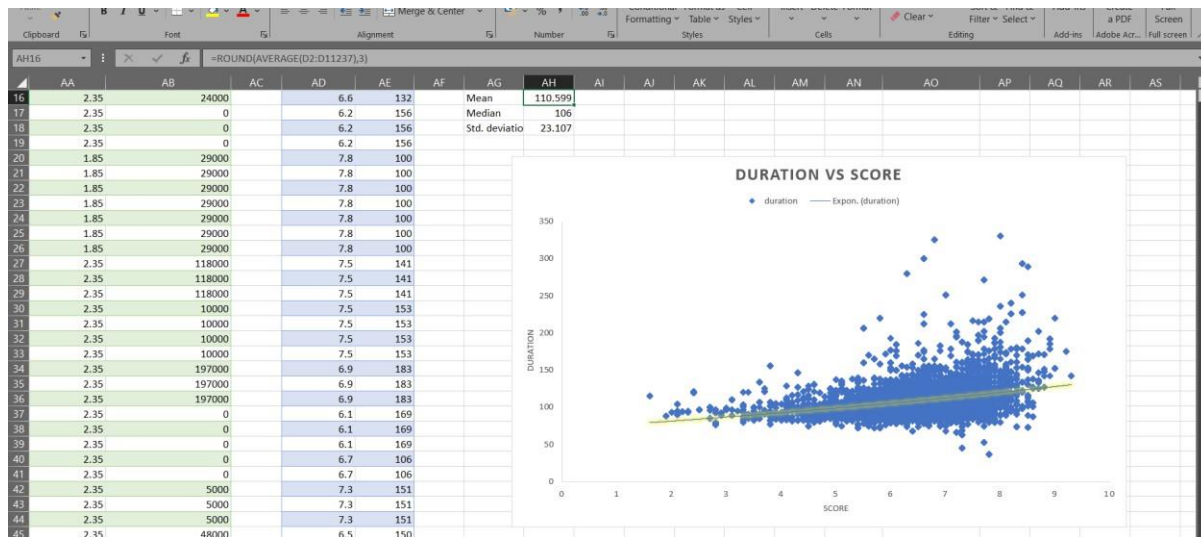
Through this project, I improved my Excel querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends in hiring in different gender, and position tiers. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

- **Movie Genre Analysis:**

Clipboard		Font		Alignment		Number		Formatting		Table		Styles		Cells		Editing		Add-ins		Adobe Acro...		Full screen	
A03																							
=ROUND(STDEV.S(IF(\$J\$2:\$J\$11237=AH3,\$Z\$2:\$Z\$11237)),3))																							
	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP					
1	actor_2_facebook_likes	imdb_score	aspect_ratio	movie_facebook_likes					S.No.	Genre	Count	Mean	Mode	Median	Range	Variance	Std. Deviation						
2	936	7.9	1.78	33000					1	Drama	1893	6.79	6.7	6.9	7.2	0.804	0.897						
3	936	7.9	1.78	33000					2	Comedy	1461	6.19	6.7	6.3	6.9	1.073	1.036						
4	936	7.9	1.78	33000					3	Thriller	1117	6.38	6.5	6.4	6.3	0.944	0.972						
5	936	7.9	1.78	33000					4	Action	959	6.29	6.6	6.3	6.9	1.079	1.039						
6	5000	7.1	2.35		0				5	Romance	859	6.44	6.5	6.5	6.4	0.911	0.954						
7	5000	7.1	2.35		0				6	Adventure	781	6.45	6.6	6.6	6.6	1.241	1.114						
8	5000	7.1	2.35		0				7	Crime	709	6.55	6.6	6.6	6.9	0.968	0.984						
9	393	6.8	2.35	85000					8	Fantasy	507	6.28	6.7	6.4	6.7	1.286	1.134						
10	393	6.8	2.35	85000					9	Sci-Fi	496	6.33	6.7	6.4	6.9	1.348	1.161						
11	393	6.8	2.35	85000					10	Family	442	6.21	5.4	6.3	6.7	1.35	1.162						
12	23000	8.5	2.35	164000																			
13	23000	8.5	2.35	164000																			
14	632	6.6	2.35	24000																			
15	632	6.6	2.35	24000																			
16	632	6.6	2.35	24000																			
17	11000	6.2	2.35	0																			
18	11000	6.2	2.35	0																			

• Movie Duration Analysis:



• Language Analysis:

Clipboard		Font		Alignment		Number		Styles		Cells		Editing		Add-ins	
AJ2															
=ROUND(STDEV.S(IF(\$T\$2:\$T\$11237=AF2,\$Z\$2:\$Z\$11237)),3))															
	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN
1	imdb_score	aspect_ratio	movie_facebook_likes			S.NO.	Language	Count	Mean	Median	Std. Deviation				
2	7.9	1.78	33000			1	English	10791	6.42	6.5	1.037				
3	7.9	1.78	33000			2	French	97	7.4	7.3	0.479				
4	7.9	1.78	33000			3	Spanish	59	7.12	7.2	0.849				
5	7.9	1.78	33000			4	Mandarin	51	7.06	7.4	0.754				
6	7.1	2.35	0			5	Japanese	37	7.59	7.9	0.947				
7	7.1	2.35	0			6	German	28	7.8	7.8	0.638				
8	7.1	2.35	0			7	Cantonese	21	7.38	7.3	0.311				
9	6.8	2.35	85000			8	Korean	16	7.63	7.5	0.546				
10	6.8	2.35	85000			9	Italian	15	6.81	6.9	1.162				
11	6.8	2.35	85000			10	Hindi	11	7.42	7.4	0.621				
12	8.5	2.35	164000												
13	8.5	2.35	164000												
14	6.6	2.35	24000												

• Director Analysis:

	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK
1	actor_facebook_likes	imdb_score	aspect_ratio	movie_facebook_likes					Director	Average of imdb_score	Percentile		
2	936	7.9	1.78	33000					Akira Kurosawa	8.70	Top 1%		
3	936	7.9	1.78	33000					Charles Chaplin	8.60	Top 1%		
4	936	7.9	1.78	33000					Tony Kaye	8.60	Top 1%		
5	936	7.9	1.78	33000					Alfred Hitchcock	8.50	Top 1%		
6	5000	7.1	2.35	0					Damien Chazelle	8.50	Top 1%		
7	5000	7.1	2.35	0					Majid Majidi	8.50	Top 1%		
8	5000	7.1	2.35	0					Ron Fricke	8.50	Top 1%		
9	393	6.8	2.35	85000					Christopher Nolan	8.47	Top 1%		
10	393	6.8	2.35	85000		8.205	99		Asghar Farhadi	8.40	Top 1%		
11	393	6.8	2.35	85000					Richard Marquand	8.40	Top 1%		
12	23000	8.5	2.35	164000					Billy Wilder	8.30	Top 1%		
13	23000	8.5	2.35	164000					Fritz Lang	8.30	Top 1%		
14	632	6.6	2.35	24000		7.5	90		Lee Unkrich	8.30	Top 1%		
15	632	6.6	2.35	24000		7.7	95		Lenny Abrahamson	8.30	Top 1%		
16	632	6.6	2.35	24000					Sergio Leone	8.28	Top 1%		
17	11000	6.2	2.35	0		7	75		Pete Docter	8.23	Top 1%		
18	11000	6.2	2.35	0					Hayao Miyazaki	8.21	Top 1%		
19	11000	6.2	2.35	0					Ella Kazan	8.20	Top 5%		
20	553	7.8	1.85	29000					Joshua Oppenheimer	8.20	Top 5%		
21	553	7.8	1.85	29000					Juan José Campanella	8.20	Top 5%		
22	553	7.8	1.85	29000					George Roy Hill	8.19	Top 5%		
23	553	7.8	1.85	29000					Frank Darabont	8.18	Top 5%		
24	553	7.8	1.85	29000					Quentin Tarantino	8.16	Top 5%		
25	553	7.8	1.85	29000					Victor Fleming	8.15	Top 5%		
26	553	7.8	1.85	29000					David Singleton	8.10	Top 5%		
27	21000	7.5	2.35	118000					Je-kyu Kang	8.10	Top 5%		
28	21000	7.5	2.35	118000					Michael Wadleigh	8.10	Top 5%		
29	21000	7.5	2.35	118000					Terry George	8.10	Top 5%		
30	11000	7.5	2.35	10000					Tim Miller	8.10	Top 5%		
31	11000	7.5	2.35	10000					William Wyler	8.10	Top 5%		

- Budget Analysis:**

Clipboard Font Alignment Number Styles Cells Editing Add-ins													
=IF(ROUND((((I2-AC2)/AC2)*100),3)<0,0,ROUND((((I2-AC2)/AC2)*100),3))													
	Z	AA	AB	AC	AD	AE	AF	AG	Formula Bar	AH	AI	AJ	AK
1	imdb_score	aspect_ratio	movie_facebook_likes	Budget				S.No.	Movie		Profit %	Correlation	
2	7.9	1.78	33000	237000000				1	Paranormal Activity		220.889	0.154982894	
3	7.1	2.35	0	300000000				2	Tarnation		3.135	0.154982894	
4	6.8	2.35	85000	245000000				3	The Blair Witch Project		-18.337	0.154982894	
5	8.5	2.35	164000	250000000				4	The Brothers McMullen		79.252	0.154982894	
6	6.6	2.35	24000	263700000				5	The Texas Chain Saw Massacre		-72.295	0.154982894	
7	6.2	2.35	0	258000000				6	El Mariachi		30.438	0.154982894	
8	7.8	1.85	29000	260000000				7	The Gallows		-22.766	0.154982894	
9	7.5	2.35	118000	250000000				8	Super Size Me		83.597	0.154982894	
10	7.5	2.35	10000	250000000				9	Halloween		20.783	0.154982894	
11	6.9	2.35	197000	250000000				10	American Graffiti		32.1	0.154982894	
12	6.1	2.35	0	209000000		Highest Grossing		11	Rocky		-4.273	0.154982894	
13	6.7	2.35	0	200000000		Good Kill		12	In the Company of Men		-15.816	0.154982894	
14	7.3	2.35	5000	225000000				13	Napoleon Dynamite		88.015	0.154982894	
15	6.5	2.35	48000	215000000				14	Facing the Giants		-58.47	0.154982894	
16	7.2	2.35	118000	225000000				15	Snow White and the Seven Dwarfs		29.343	0.154982894	
17	6.6	2.35	0	225000000				16	Benji		-37.06	0.154982894	
18	8.1	1.85	123000	220000000				17	My Date with Drew		183.309	0.154982894	
19	6.7	2.35	58000	250000000				18	The Circle		-3.574	0.154982894	
20	6.8	1.85	40000	225000000				19	Fireproof		-20.435	0.154982894	
21	7.5	2.35	65000	250000000				20	Open Water		2.043	0.154982894	
22	7	2.35	56000	230000000				21	Primer		13.926	0.154982894	
23	6.7	2.35	17000	200000000				22	Pi		-47.39	0.154982894	
24	7.9	2.35	83000	225000000				23	Slacker		14.825	0.154982894	
25	6.1	2.35	0	180000000				24	The Devil Inside		-61.065	0.154982894	
26	7.2	2.35	0	207000000				25	Once		5.339	0.154982894	
27	7.7	2.35	26000	200000000				26	Hollywood Shuffle		229.336	0.154982894	
28	8.2	2.35	72000	250000000				27	Gone with the Wind		62.879	0.154982894	
29	5.9	2.35	44000	209000000				28	My Big Fat Greek Wedding		-68.817	0.154982894	
30	7	2	150000	150000000				29	Chasing Amy		334.785	0.154982894	
31	7.8	2.35	80000	200000000				30	Animal House		52.18	0.154982894	

Project 6- Bank Loan Case Study

Overview

In this project, as a data analyst, I am given a dataset related to bank loan cases. My job is to analyse this data and answer certain questions that can help stakeholders make informed decisions.

Here's what I will be doing:

1. **Handling Missing Data:** Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. **Clubbing Columns:** If there are columns with multiple categories that can be combined, do so to simplify your analysis.
3. **Outlier Detection:** Check for outliers in the dataset that may skew your analysis.
4. **Removing Outliers:** Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. **Data Summary:** After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings and data types etc... , exploring dimensions, target variable, loan,

income, status etc. then thought about main goal of this analysis which is majorly around bank loan vs income of the individual.

Secondly, I created the required dataset from given dataset using various excel functions.

Later, I wrote excel commands for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

Here is the link of my project file :-

https://docs.google.com/spreadsheets/d/1fe9VJqwt75AvI0l22Uf5r4Dn32Ws7tK/edit?usp=drive_link&ouid=107716378045740610577&rtpof=true&sd=true

Insights

- **Identify Missing Data :** there are 14,88,212 blanks in application dataset, I replaced those with average values of those columns.
- **Outlier Analysis:** these indicate people take very high loans as compared to their income ,suggesting higher default risk.
- **Data Imbalance Analysis:** there is a high approval rate but one third of applications do not results in loans , this needs to be improved.
- **Univariate, Segmented Univariate, and Bivariate Analysis:** Higher income individuals have higher loan approval rate.
- **Correlation :** Higher income does **not strongly influence loan size**. Banks are not scaling loan amounts significantly with income. Region seems to be a slightly better factor in explaining default rate than income or loan.

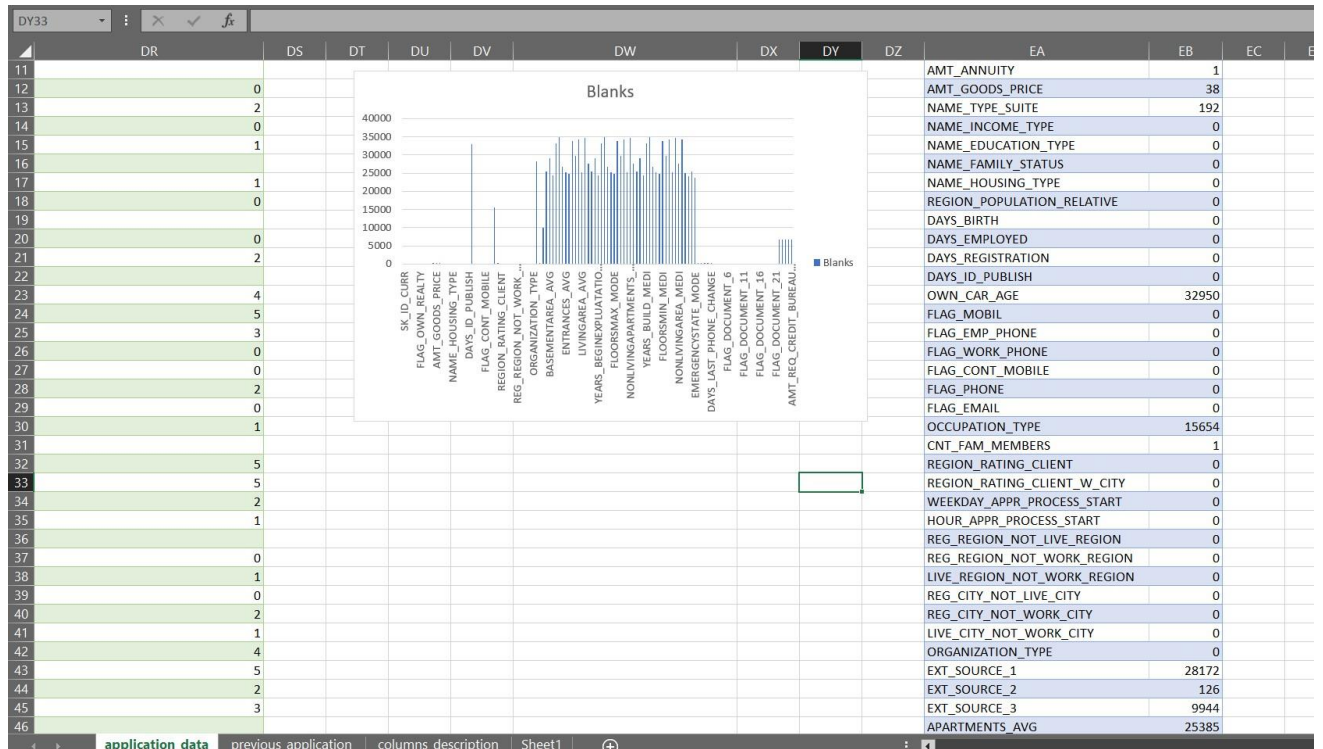
Results

Through this project, I improved my Excel querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends in giving loan to various individuals and default rate based on their profession, income, region etc. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear ,

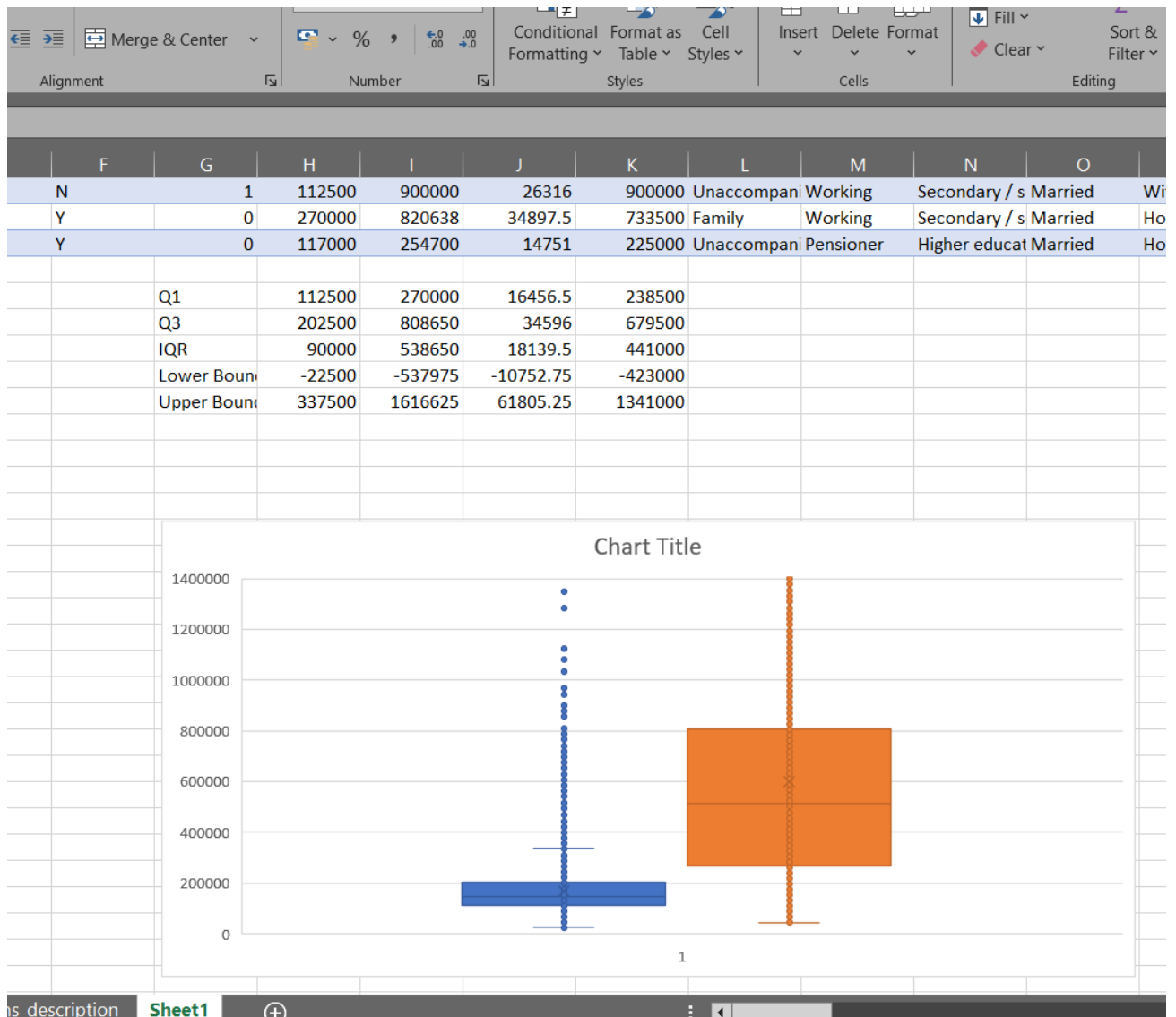
structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

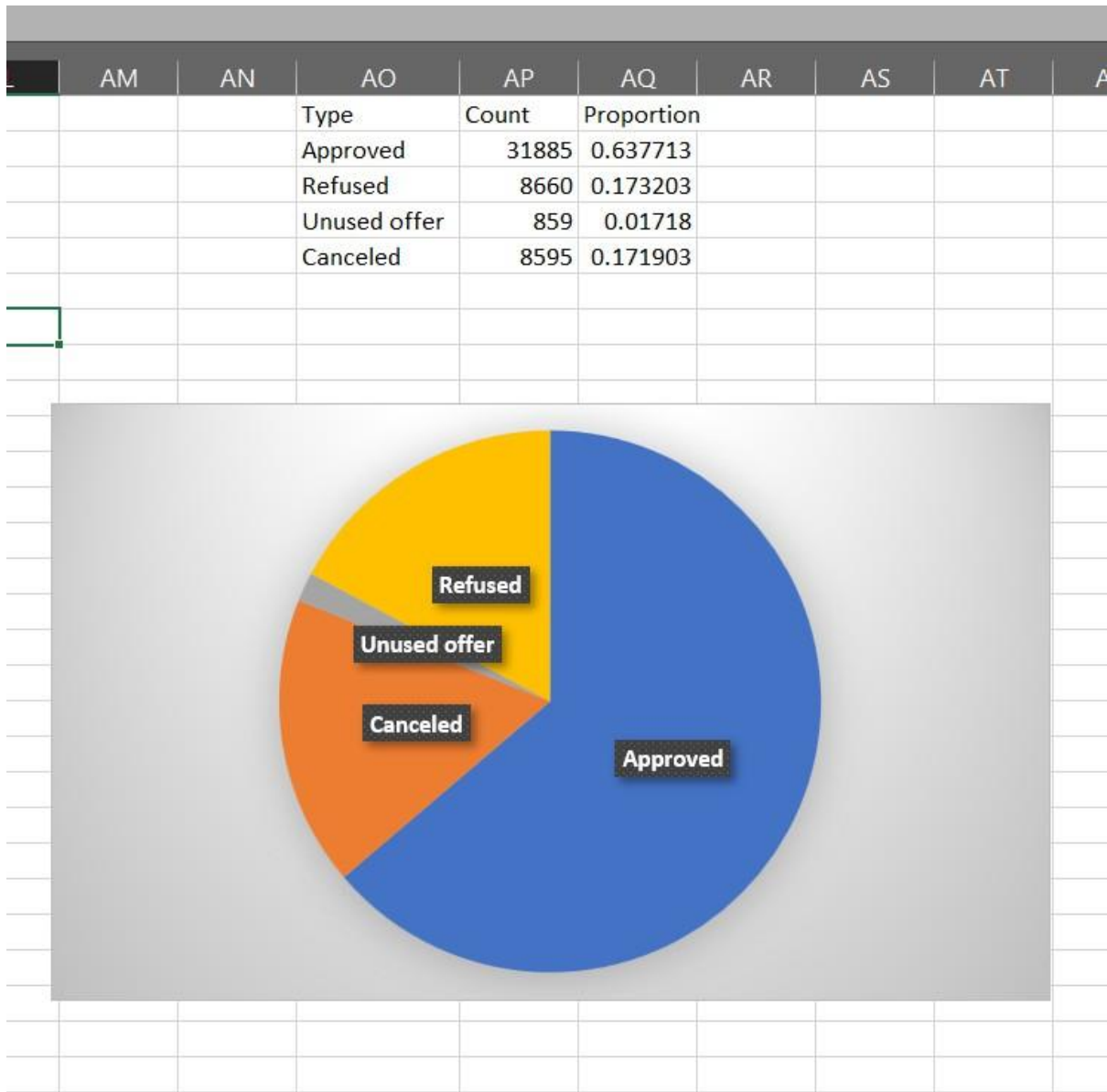
- **Identify missing data:**



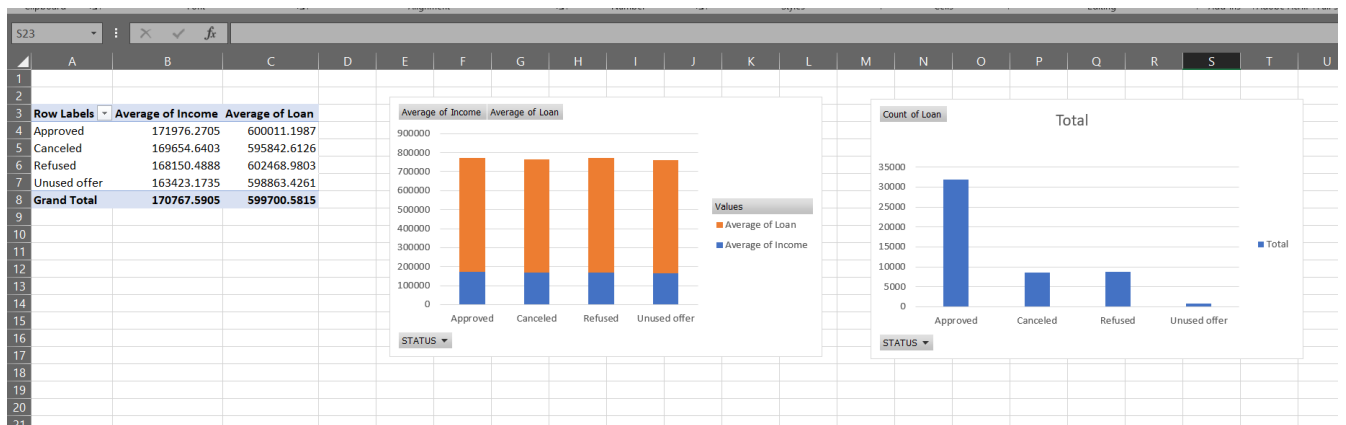
- **Outlier Analysis :**



- **Data Imbalance analysis :**



- **Univariate, Segmented Univariate, and Bivariate Analysis:**



- Correlation :

	A	B	C	D	E
1		Income	Loan	Target	Region
2	Income	1			
3	Loan	0.069316	1		
4	Target	0.010894	-0.03243	1	
5	Region	-0.03819	-0.10051	0.06613	1
6					

	A	B	C	D	E	F	G	H	I	J	K	L
1	Income	Loan	Target	Region	STATUS	0	1					
2	202500	406597.5	1	2	Approved	2	1					
3	270000	1293503	0	1	Approved	1	2			Mean	Median	STD. DEV.
4	67500	135000	0	2	Approved	1	2		Income	170767.6	145800	531819.1
5	135000	312682.5	0	2	Approved	1	2		Loan	599700.6	514777.5	402415.4
6	121500	513000	0	2	Refused	1	2					
7	99000	490495.5	0	2	Approved	1	2					
8	171000	1560726	0	2	Canceled	1	2					
9	360000	1530000	0	3	Canceled	1	2					
10	112500	1019610	0	2	Canceled	1	2					
11	135000	405000	0	2	Canceled	1	2					
12	112500	652500	0	2	Approved	1	2					
13	38419.16	148365	0	2	Approved	1	2					4500000
14	67500	80865	0	2	Approved	1	2					4000000
15	225000	918468	0	2	Approved	1	2					3500000
16	189000	773680.5	0	2	Approved	1	2					3000000
17	157500	299772	0	3	Approved	1	2					2500000
18	108000	509602.5	0	2	Approved	1	2					2000000
19	81000	270000	0	2	Approved	1	2					1500000
20	112500	157500	0	1	Approved	1	2					1000000
21	90000	544491	0	2	Refused	1	2					500000
22	135000	427500	0	2	Refused	1	2					0
23	202500	1132574	0	2	Approved	1	2					
24	450000	497520	0	3	Refused	1	2					
25	83250	239850	0	3	Refused	1	2					
26	135000	247500	0	2	Canceled	1	2					
27	90000	225000	0	2	Approved	1	2					
28	112500	979992	1	3	Approved	2	1		Target	0	1	
29	112500	327024	0	2	Approved	1	2		Income	0.010894	-0.01089	
30	270000	790830	0	1	Approved	1	2		Loan	-0.03243	0.032428	
31	90000	180000	0	2	Refused	1	2		Region	0.06613	-0.06613	

Chart Area

Project 7- Analyzing the Impact of Car Features on Price and Profitability

Link:

https://docs.google.com/spreadsheets/d/1ydgNUPwofUOAbmomb4OF9oBqrwZZ-wT/edit?usp=drive_link&ouid=107716378045740610577&rtpof=true&sd=true

Overview

In this project, as a data analyst, I am given a dataset related to various cars and its features. My job is to analyse this data and answer certain questions that can help stakeholders make informed decisions.

Here's what I will be doing:

1. **Handling Missing Data:** Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. **Clubbing Columns:** If there are columns with multiple categories that can be combined, do so to simplify your analysis. In this dataset, make and model are the ones.
3. **Outlier Detection:** Check for outliers in the dataset that may skew your analysis.
4. **Removing Outliers:** Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. **Data Summary:** After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings, and data types etc... , exploring dimensions, car's brand, model, mileage in miles per gallons etc. then thought about main goal of this analysis which is majorly around how can a car manufacturer optimize pricing and product development decisions to maximize profitability while meeting consumer demand ?

Secondly, I created the required dataset from given dataset using various excel functions.

Later, I wrote excel commands for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

Insights

- **Popularity of a car model in different market categories:** Diesel cars are the most popular per brand and Luxury cars have the largest count but lowest popularity.
- **Car's engine vs price :** As engine horsepower **increases**, the MSRP (price) also increases exponentially . Mainstream market zone is around 400-700 HP .
- **Price dependency on various features :** Car's price is most strongly driven by performance (horsepower), followed by highway efficiency. Meanwhile, practical features like more doors or high city mileage are associated with lower prices, since luxury/performance cars are less practical but more expensive.
- **Average price of a car across different manufacturers:** Car pricing is heavily brand-driven. From affordable Budget cars to Premium cars segment and highly expensive Luxury Cars.
- **Relationship between fuel efficiency and the number of cylinders :** Number of cylinders **is not a reliable predictor of mileage** in this dataset.
- **Dashboard insights :**

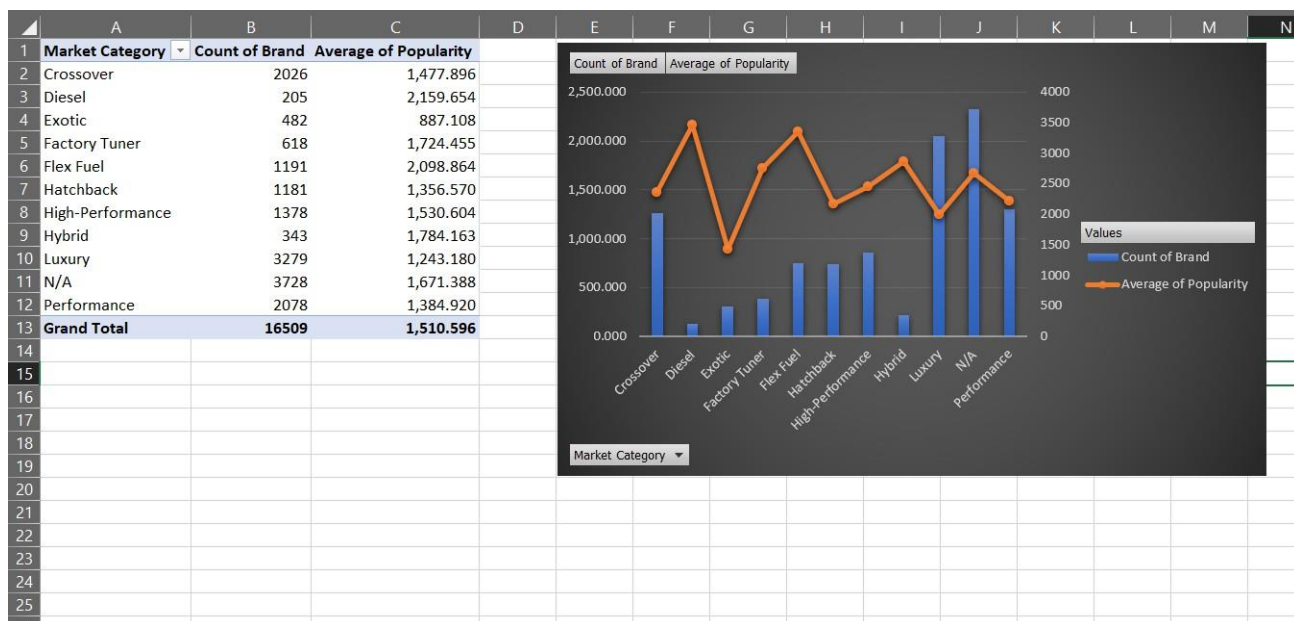
1. Luxury brands dominate average pricing, while mass-market brands stay in affordable ranges.
2. Mass-market brands generate more total revenue than niche luxury brands due to scale.
3. Fuel economy trades off with performance. Supercars are clear outliers.
4. Transmission type is not a strong price driver — brand and luxury level dominate.
5. Body style has some correlation with transmission type.
6. Fuel efficiency has improved over time, especially for smaller cars. SUVs and vans lag behind .

Results

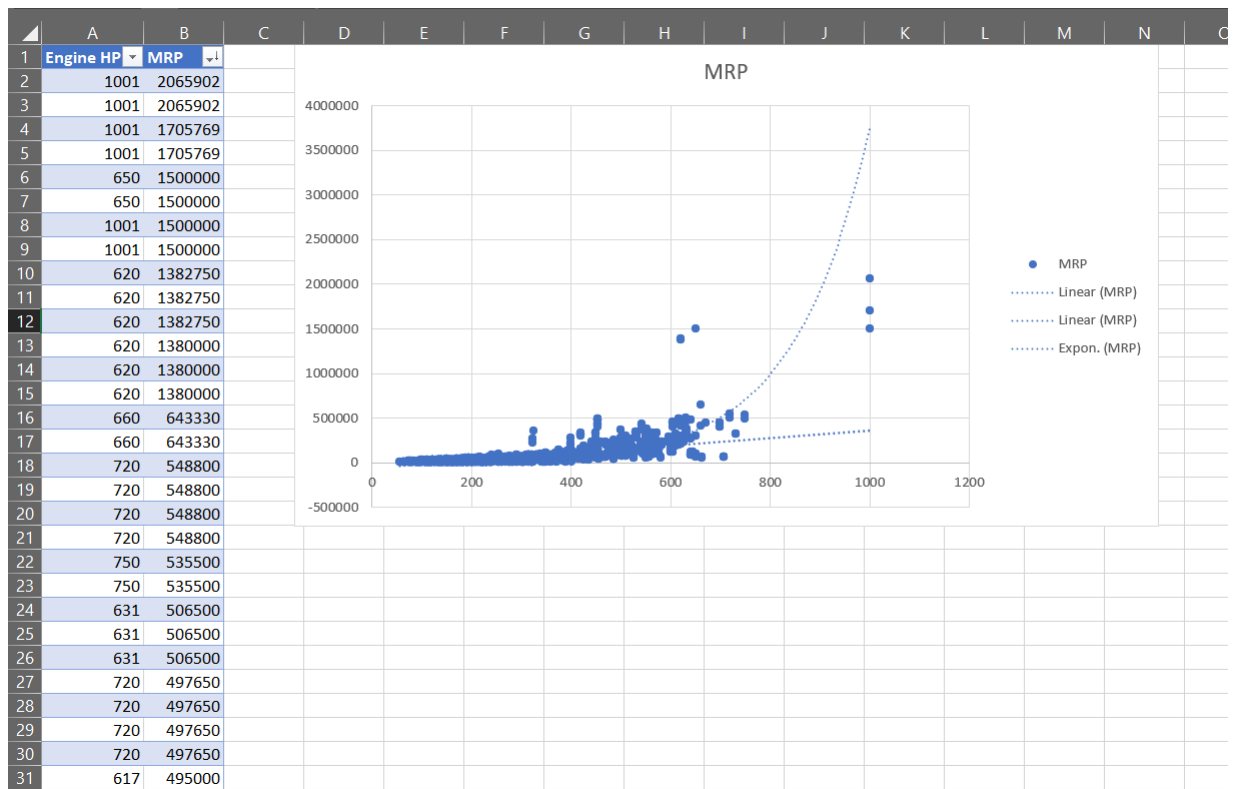
Through this project, I improved my Excel querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends of car features vs its popularity and pricing. It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

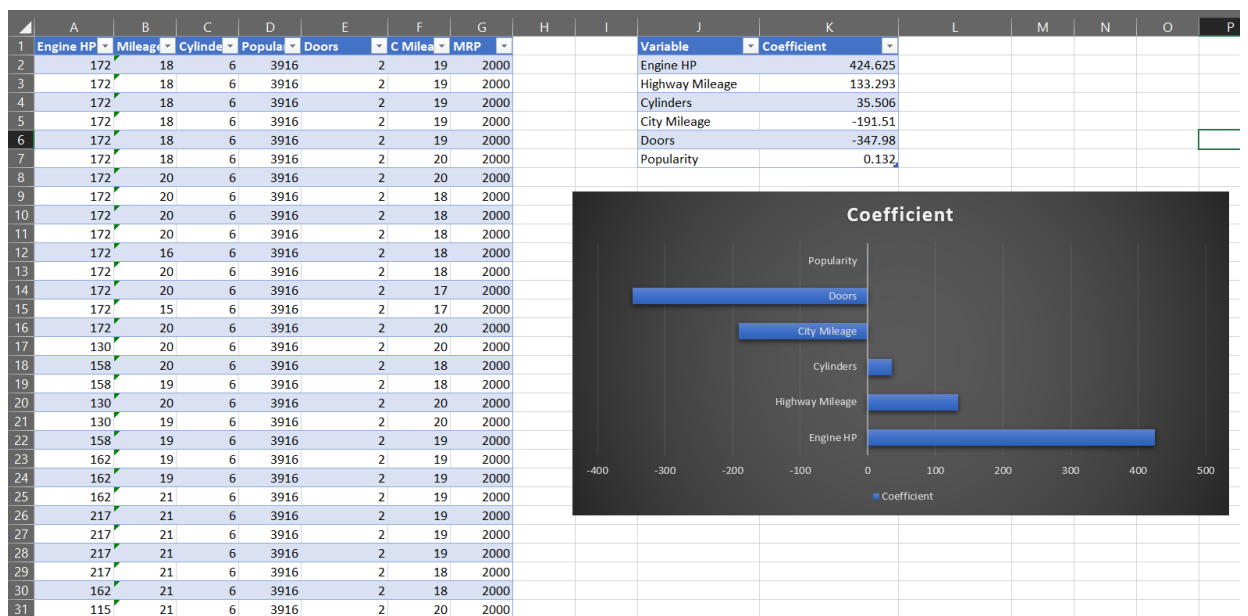
- **Popularity of a car model in different market categories**



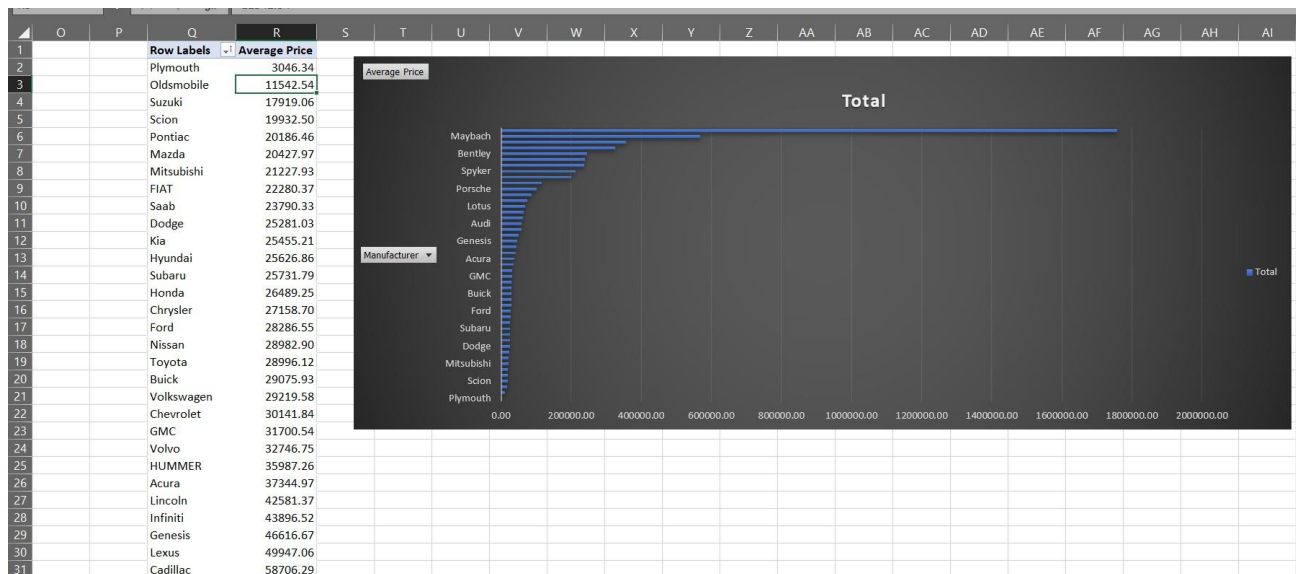
- **Car's engine vs price**



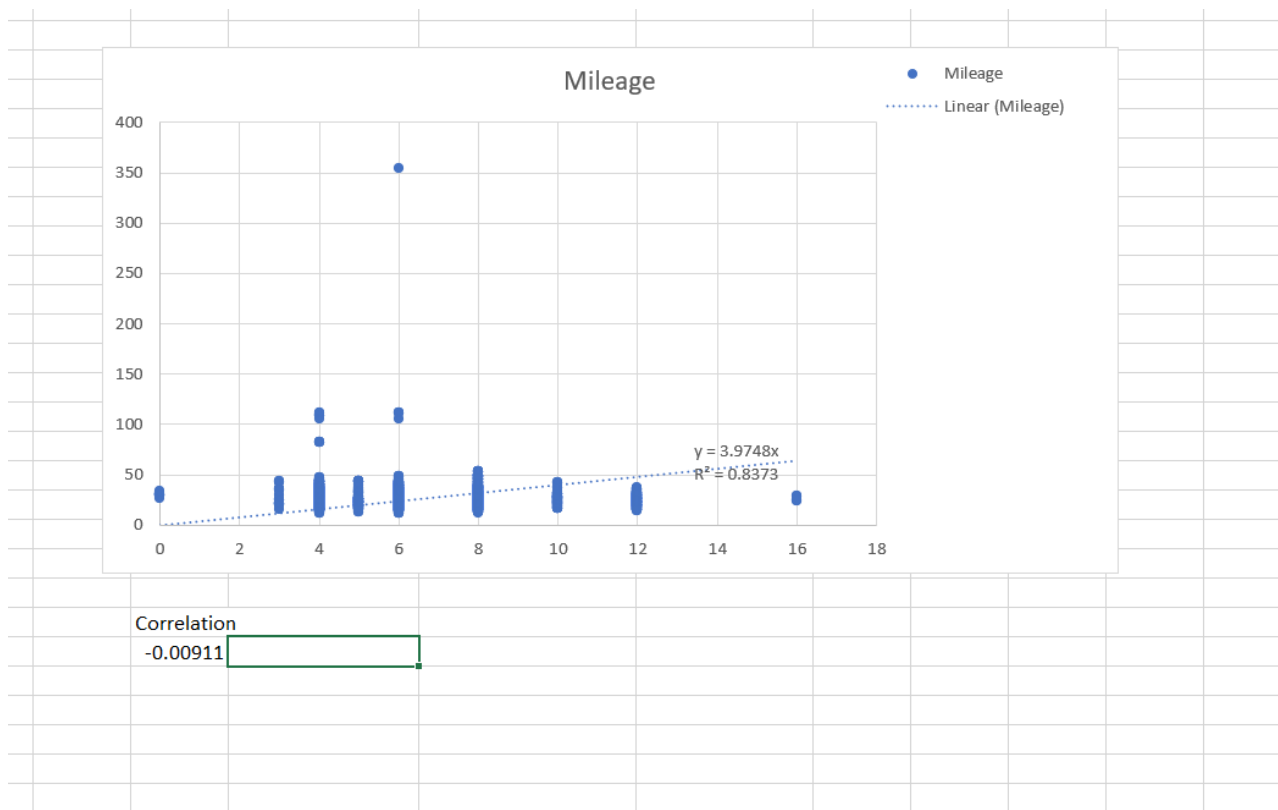
- Price dependency on various features



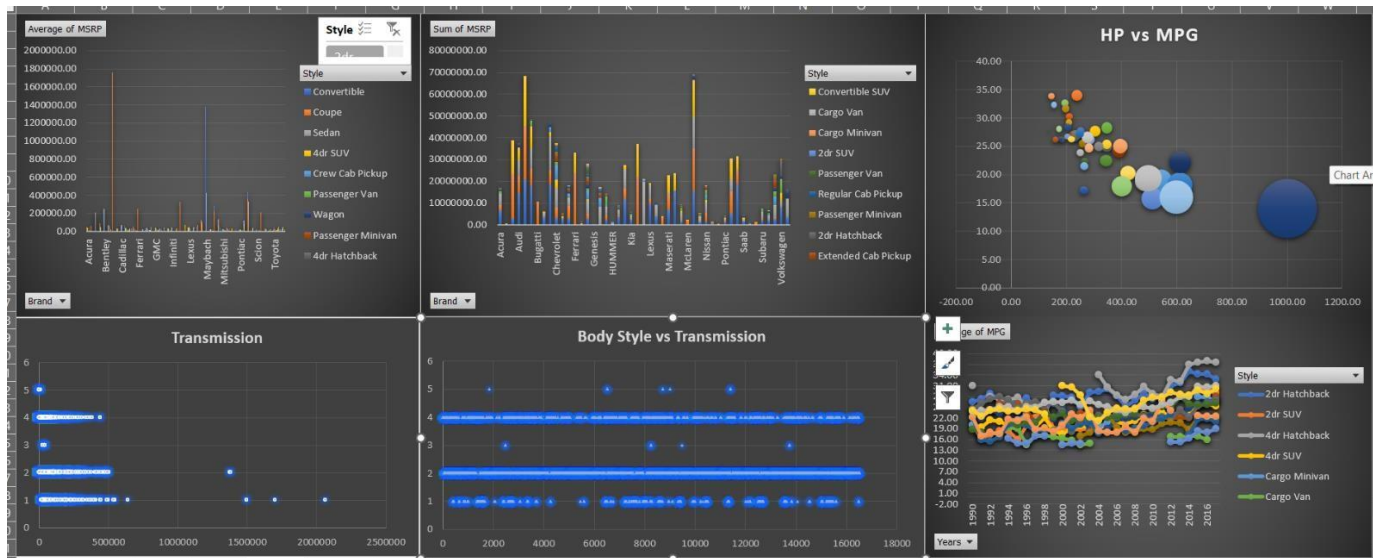
- Average price of a car across different manufacturers



- Relationship between fuel efficiency and the number of cylinders



- Dashboard



Project 8- ABC Call Volume Trend Analysis

Link:

https://docs.google.com/spreadsheets/d/12kLBv0ZLcFN5AKm7z70GQ_yIWuCv5kVx/edit?usp=drive_link&oid=107716378045740610577&rtpof=true&sd=true

Overview

In this project, as a data analyst, I am given a dataset related to inbound calling team of a company. My job is to analyse this data and answer certain questions that can help stakeholders make informed decisions.

Here's what I will be doing:

1. **Handling Missing Data:** Check if there are any missing values in the dataset. If there are, decide on the best strategy to handle them.
2. **Clubbing Columns:** If there are columns with multiple categories that can be combined, do so to simplify your analysis.
3. **Outlier Detection:** Check for outliers in the dataset that may skew your analysis.
4. **Removing Outliers:** Decide on the best strategy to handle outliers. This could be removing them, replacing them, or leaving them as is, depending on the situation.
5. **Data Summary:** After cleaning and preparing your data, summarize your findings. This could involve calculating averages, medians, or other statistical measures. It could also involve creating visualizations to better understand the data.

My approach will be structured and insight driven.

Approach

Firstly, I got familiar with dataset , identified the tables and columns, column meanings, and data types etc... , exploring dimensions, call timings, time spent, agent's name and id etc. then thought about main goal of this analysis which is majorly around average number of calls, number of agents required per hour and average time spent.

Secondly, I created the required dataset from given dataset using various excel functions.

Later, I wrote excel commands for all questions asked by the teams and summarized my insights .

Software

I have used MS Excel , since it makes handling large dataset easy and is easily integrable with relational databases.

Insights

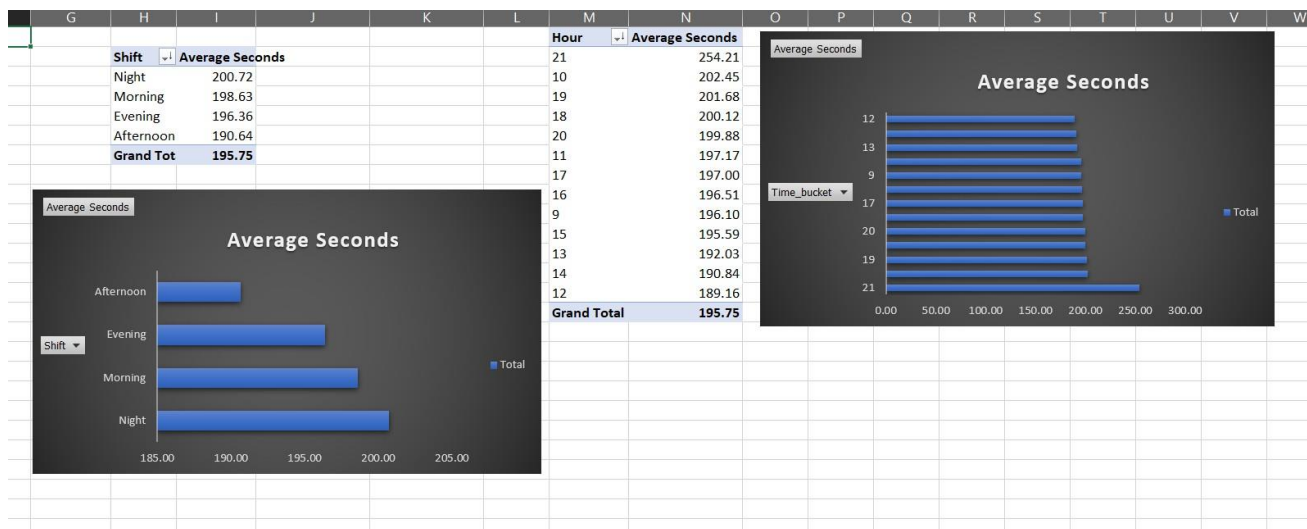
- **Average Call Duration:** The overall grand average is **195.75 seconds**, and most shifts/hours stay close to this number. Except 9 to 10 pm shift indicating requires more agents or better process optimization.
- **Call Volume Analysis :** The highest demand is between 12 PM and 5 PM, requiring maximum staffing and resources.
- **Manpower Planning :**
 1. The busiest hours are 10 AM to 1 PM, with the highest answered call volume and agent requirement. For example, at 11 AM, you need ~1203 agents.
 2. Unplanned leave is adding ~**15–17% extra staffing requirement**. If this trend continues, staffing during peak hours (10–1 PM) becomes a **critical risk area**, needing up to **1300+ agents** instead of the planned 1100–1200.
- **Night Shift Manpower Planning :** The 5–8 AM window is the **critical pressure point** without buffer planning; service levels will collapse just before day shift handover.

Results

Through this project, I improved my Excel querying skills and developed a strong understanding of real-world data analysis. I have successfully identified key trends of call duration , volume vs required agents . It also helped me to enhance my ability to extract actionable insights from large datasets and present them in a clear , structured way , it was a golden opportunity to gain experience as data analyst role.

Below are the required results:

- **Average Call Duration**



- **Call Volume Analysis**

	Total night calls>>	46169	Avg minutes	Agents required	Due to unplanned leave	
	22	4617	3.63	419	482	
	23	4617	3.74	432	497	
	0	3093	3.66	283	325	
	1	3093	3.5	271	311	
	2	1524	3.56	136	156	
	3	1524	3.53	134	154	
	4	3093	3.62	280	322	
	5	4617	3.64	420	483	
	6	6140	3.64	559	643	
	7	6140	3.71	569	654	
	8	7711	3.73	719	826	
						

Learnings

Diving into these eight projects was like travelling various cities of India ,each place with its own culture, people, and thrilling adventures. But hey, that is how you learn, right?

Data Analytics Process taught me the discipline of structuring data work. It is like learning how to plan everything for a trip as you cannot enjoy the adventure if there are a lot of loose ends. Breaking problems into steps, asking the right questions, and documenting findings became my secret weapons.

Instagram User Analytics was basically watching netizens with SQL . From engagement rates to follower behaviour, I realized data tells us far more than filters and hashtags ever could. Lesson learned: people may lie, but numbers rarely do (unless you forget a join).

With Operation & Metric Analytics, I felt like a detective in an efficiency crime scene. Where was time leaking? Which process slowed things down? I discovered the joy of KPIs and dashboards, and the pain of forgetting to update them. Numbers became less scary when I saw how they could actually save

Hiring Process Analytics was eye-opening. While going through recruitment data showed me how companies can unintentionally create salary disparities . Excel formulas became my flashlight, highlighting inefficiencies in hiring pipelines. Also, I learned recruiters love pie charts almost as much as candidates love ghosting.

The IMDB Movie Analysis project was the most fun—finally, binge-watching felt like research. Ratings, genres, and box office trends taught me how storytelling works both in movies and in data. And yes, SQL can tell you why rom-coms are not top rated.

Bank Loan Case Study was a reality check. Behind every row of data is someone hoping for financial support. It sharpened my ability to identify risk, understand patterns in defaults, and appreciate how tiny decimal errors can create giant consequences.

Impact of Car Features was like playing with Lego, mixing variables, testing which features matter, and realizing horsepower is cool but not always king. Excel regression tools became less intimidating, and I finally understood why my friend's cheap hatchback outsells flashy sedans.

Finally, ABC Call Volume Trend reminded me patience is a skill. It was the nail in coffin, but the journey showed me how tricky time-series data can be. Peaks, troughs, and call centre chaos taught me the value of smoothing data, because life rarely follows straight lines. Overall, these projects made me less afraid of messy spreadsheets and more confident with SQL queries. Data stopped being just “rows and columns” and started feeling like un amigo raro ,you never fully understand them, but you are glad they are around.